



Pacific Hills

Wisdom & Knowledge in Christ

Senior School Examination

2016

HSC TRIAL
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time- 5 minutes
- Working Time - 3 hours
- Write using a blue or black pen
- Board approved calculators may be used
- A Standard Integrals Sheet is provided at the back of this paper which may be detached and used throughout the paper.
- Marks may be deducted for careless, untidy, or badly arranged work

Total Marks 100

Section I

10 marks

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided. This sheet should be detached.
- Allow about 15 minutes for this section.

Section II

90 marks

- Attempt questions 11 – 16
- Answer in the booklets provided. Start a new booklet for each question.
Allow about 2 hours & 45 minutes

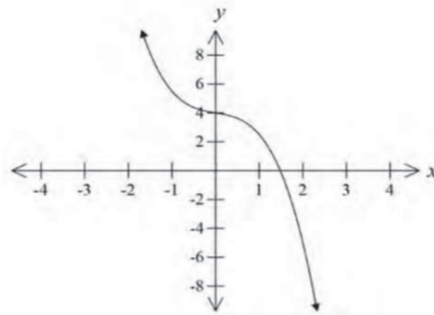
Student Number: _____

Section 1**10 marks Attempt Questions 1 – 10****Allow about 15 minutes for this section** Use the multiple-choice answer sheet

- 1 An object rotates at 40 rpm and is moving at 30 m/s. The radius of the motion is
- (A) 1.33 m
- (B) 6.37 m
- (C) 7.16 m
- (D) 20m
- 2 The eccentricity of the hyperbola $4x^2 - 25y^2 = 9$ is?
- (A) $\frac{\sqrt{21}}{5}$
- (B) $\frac{\sqrt{29}}{5}$
- (C) $\frac{\sqrt{21}}{2}$
- (D) $\frac{\sqrt{29}}{2}$
- 3 Let $z = 3 - i$. What is the value of $\overline{iz}$?
- (A) $-1 - 3i$
- (B) $-1 + 3i$
- (C) $1 - 3i$
- (D) $1 + 3i$.

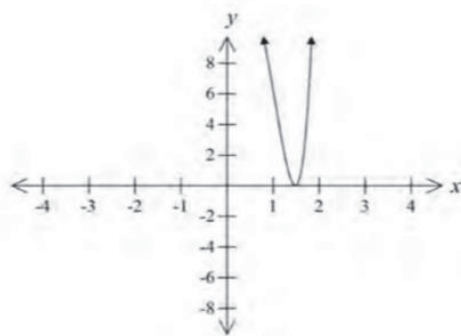
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The diagram below shows the graph of the function $y = f(x)$.

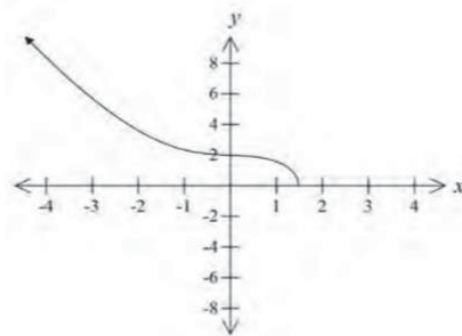


Which diagram represents the graph of $y^2 = f(x)$?

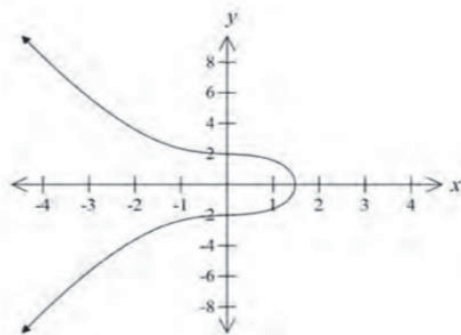
(A)



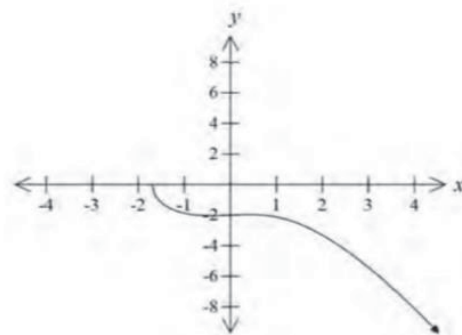
(B)



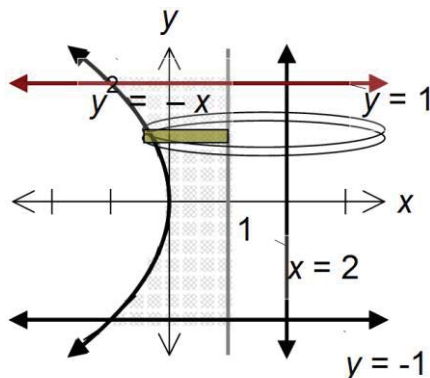
(C)



(D)



- 5 The region bounded by the lines $x = 1$, $y = 1$, $y = -1$ and the curve $x = -y^2$.
The region is rotated through 360° about the line $x = 2$ to form a solid.
What is the correct expression for the volume of the solid?



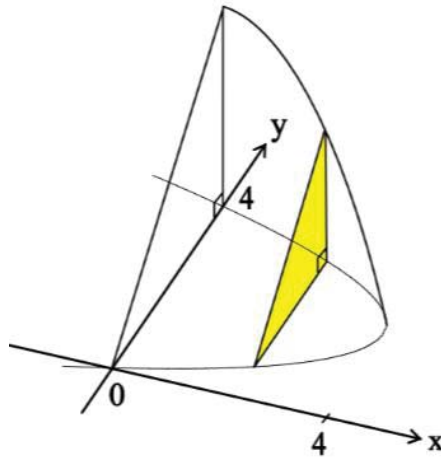
- $$\begin{aligned} \text{(A)} \quad V &= \int_{-1}^1 \pi(y^4 - 4y^2 + 3) dy \\ \text{(B)} \quad V &= \int_{-1}^1 \pi(y^4 + 4y^2 + 3) dy \\ \text{(C)} \quad V &= \int_{-1}^1 \pi(y^4 - 4y^2 + 4) dy \\ \text{(D)} \quad V &= \int_{-1}^1 \pi(y^4 + 4y^2 + 4) dy \end{aligned}$$

- 6 A particle of mass m falls from rest under gravity and the resistance to the motion is mkv^2 where v is its speed and k is a positive constant. Which of the following is the correct expression for the square of the velocity, where x is the distance fallen?

- $$\begin{aligned} \text{(A)} \quad & v^2 = \frac{g}{k} (1 - e^{-2kx}) \\ \text{(B)} \quad & v^2 = \frac{g}{k} (1 + e^{-2kx}) \\ \text{(C)} \quad & v^2 = \frac{g}{k} (1 - e^{2kx}) \\ \text{(D)} \quad & v^2 = \frac{g}{k} (1 + e^{2kx}) \end{aligned}$$

- 7 Which of these ellipses has foci $(0, \pm 3)$?
- (A) $8x^2 + y^2 = 8$
- (B) $5x^2 + 4y^2 = 20$
- (C) $16x^2 + 25y^2 = 400$
- (D) $25x^2 + 16y^2 = 400$
- 8 The polynomial equation $x^3 - 2x^2 + 3 = 0$ has roots α , β , and γ
What is the value of $\alpha^3 + \beta^3 + \gamma^3$?
- (A) -2
- (B) -1
- (C) -8
- (D) 8

- 9 The base of a solid is the region bounded by the parabola $x = 4y - y^2$ and the y axis. Vertical cross sections are right angled isosceles triangles perpendicular to the x axis as shown.



Which integral represents the volume of this solid?

- (A) $\int_0^4 2\sqrt{4-x} \, dx$
- (B) $\int_0^4 \pi(4-x) \, dx$
- (C) $\int_0^4 (8-2x) \, dx$
- (D) $\int_0^4 (16-4x) \, dx$
- 10 The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ lie on the same branch of the hyperbola $xy = c^2$ ($p \neq q$). The tangents at P and Q meet at the point T . What is the equation of the normal to the hyperbola at P ?
- (A) $p^2x - py + c - cp^4 = 0$
- (B) $p^3x - py + c - cp^4 = 0$
- (C) $x + p^2y - 2c = 0$
- (D) $x + p^2y - 2cp = 0$

Section II**90 marks****Attempt Questions 11 – 16****Allow about 2 hours and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

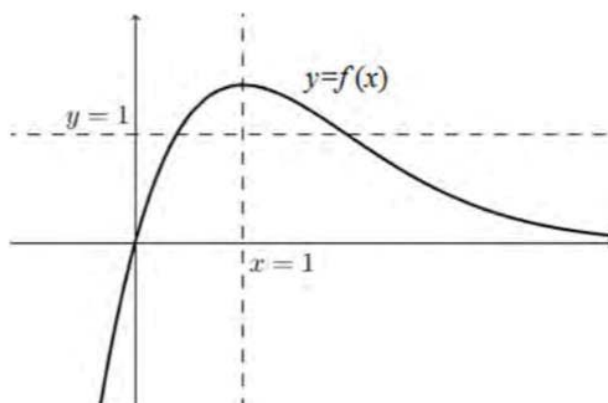
In Questions 11 – 16 your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) If $z = (1 - i)^{-1}$
- (i) Express $\bar{z}$ in modulus argument form 2
- (ii) If $(\bar{z})^{13} = a + ib$, where a and b are real numbers find the values of a and b . 2
- (b) Find the Cartesian equation of the locus of a point P which represents the complex number z where $|z - 2i| = |z|$ 2
- (c) Sketch the region in the complex plane where $\Re[(2 - 3i)z] < 12$ 2
- (d) The polynomial equation $x^3 - 3x^2 - x + 2 = 0$ has roots α , β , and γ .
Find a polynomial equation that has roots $2\alpha + \beta + \gamma$, $\alpha + 2\beta + \gamma$ and $\alpha + \beta + 2\gamma$? 3
- (e) i) Express $\frac{x^2 + x + 2}{(x^2 + 1)(x + 1)}$ in the form $\frac{Ax + B}{x^2 + 1} + \frac{C}{x + 1}$, 2
where A , B , and C are constants
- ii) Hence find $\int \frac{x^2 + x + 2}{(x^2 + 1)(x + 1)} dx$ 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a)



Using four separate graphs sketch:

i) $y = f'(x)$ 2

ii) $|y| = f(x)$ 2

iii) $y = \frac{1}{f(x)}$ 2

iv) $y = 3^{f(x)}$ 2

(b) Evaluate $\int_4^7 \frac{dx}{x^2 - 8x + 19}$ 3

(c) Let $f(x) = \frac{x^3 + 1}{x}$

i) Show that $\lim_{x \rightarrow \pm\infty} [f(x) - x^2] = 0$ 1

ii) Part (i) shows the graph of $y = f(x)$ is asymptotic to the parabola $y = x^2$.
Use this fact to help sketch the graph of $y = f(x)$. 3

Question 13 (15 marks) Use a SEPARATE writing booklet.

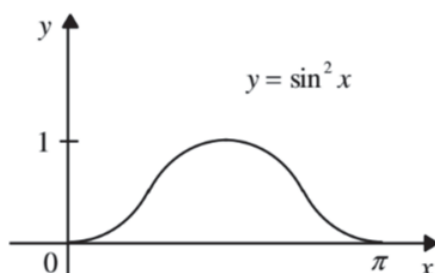
- (a) If ω is the root of $z^5 - 1 = 0$ with the smallest positive argument, find the real quadratic equation with roots $\omega + \omega^4$ and $\omega^2 + \omega^3$. **3**
- (b) Given the polynomial $P(x) = x^3 + x^2 + mx + n$ where m and n are real numbers:
- (i) If $(1 - 2i)$ is a zero of $P(x)$, factorise $P(x)$ into complex linear factors. **2**
- (ii) Find the values of m and n **2**
- (c) i) An ellipse has major and minor axes of length 12 and 8 respectively. Write a possible equation of this ellipse. **1**
- ii) A solid has the elliptical base of (i). Sections perpendicular to its base and parallel to its minor axis, are semi-circles. Find the volume of the solid **3**
- (d) i) Let $P(x)$ be a degree four polynomial with a zero of multiplicity three. Show that $P'(x)$ has a zero of multiplicity two.. **2**
- ii) Hence find all the zeros of $P(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$, given that it has a zero of multiplicity tree. **2**

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) i) Given that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$, show that $\int_0^\pi (x \cos 2x) dx = 0$

2

ii)



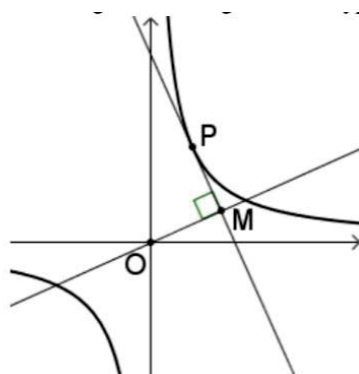
The area bounded by the curve $y = \sin^2 x$ and the x -axis between $x = 0$ and $x = \pi$ is rotated through one revolution about the y -axis.

By considering the limiting sum of the volumes of cylindrical shells find the volume of this solid

2

(b) $P\left(t, \frac{1}{t}\right)$ is a variable point on the rectangular hyperbola $xy = 1$.

M is the foot of the perpendicular from the origin to the tangent to the hyperbola at P .



i) Show that the equation of the tangent to the hyperbola at P has equation

$$x + t^2 y = 2t$$

2

ii) Find the equation of OM .

1

iii) Show that the equation of the locus of M as P varies is

$$x^4 + 2x^2 y^2 - 4xy + y^4 = 0$$

and indicate any restrictions on the values of x and y .

3

(c) A particle is fired vertically upwards with initial velocity V metres per second, and is subject to both constant gravity and air resistance proportional to speed, so that the equation of motion is given by $\ddot{x} = -g - kv$, where $k > 0$ is a constant, v is the velocity and g is acceleration due to gravity.

By using $\ddot{x} = v \frac{dv}{dx}$ and integrating prove that the projectile reaches a maximum height H

given by:
$$H = \frac{V}{k} - \frac{g}{k^2} \ln \left(1 + \frac{kV}{g} \right)$$

5

Question 15 (15 marks) Use a SEPARATE writing booklet.

(a) Use integration by parts to evaluate $\int_1^e (x^7 \log_e x) dx$ 3

(b) i) On the same diagram sketch the graphs of

$$E_1: \frac{x^2}{4} + \frac{y^2}{3} = 1 \text{ and } E_2: \frac{x^2}{16} + \frac{y^2}{12} = 1$$
2

showing clearly the intercepts on the axes.

Find the coordinates of the foci and the equations of the directrices of the ellipse E_1

ii) $P(2 \cos p, \sqrt{3} \sin p)$, where $0 < p < \frac{\pi}{2}$ is a point on ellipse E_1 .

Use differentiation to show that the tangent to the ellipse E_1 at P has equation

$$\frac{x \cos p}{2} + \frac{y \sin p}{\sqrt{3}} = 1$$
2

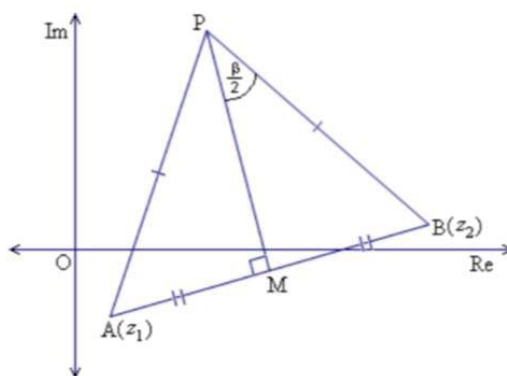
iii) The tangent to the ellipse E_1 at P meets ellipse E_2 at the points

$$Q(4 \cos q, 2\sqrt{3} \sin q) \text{ and } R(4 \cos r, 2\sqrt{3} \sin r)$$

where $-\pi < q < \pi$ and $-\pi < r < \pi$.

Show that q and r differ by $\frac{2\pi}{3}$ 2

(c)



i) Find the complex number represented by

$\alpha) \quad \overline{AM}$ 1

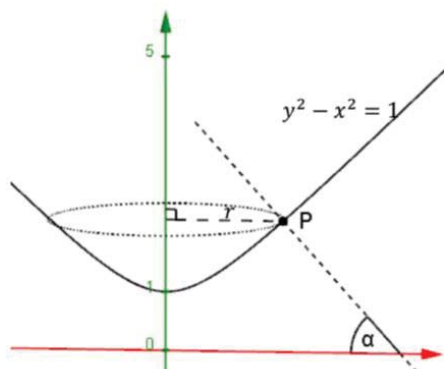
$\beta) \quad \overline{MP}$ 2

ii) Hence show that P represents the complex number

$$\frac{1}{2} \left(1 - i \cot \frac{\beta}{2} \right) z_1 + \frac{1}{2} \left(1 + i \cot \frac{\beta}{2} \right) z_2$$
3

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) A bowl is formed by rotating the hyperbola $y^2 - x^2 = 1$ for $1 \leq y \leq 5$ through 180° . Sometime later, a particle P of mass m moves around the inner surface of the bowl in a horizontal circle with constant angular velocity ω .



- (i) Show that if the radius of the circle in which P moves is r , then the normal to the surface at P makes an angle α with the horizontal as shown in the diagram where

$$\tan \alpha = \frac{\sqrt{1+r^2}}{r} . \quad 2$$

- (ii) Draw a diagram showing the forces on P . 1

- (iii) Find the expression for the radius r of the circle of motion and the magnitude of the reaction force between the surface and the particle in terms of m , g and ω . 3

- (iv) Find the values of ω for which the described motion of P is possible 1

(b) Let $I_n = \int_1^e (1 - \ln x)^n dx$ where $n = 0, 1, 2, \dots$

- (i) Show that $I_n = -1 + nI_{n-1}$ where $n = 0, 1, 2, \dots$ 2

- (ii) Hence evaluate $\int_1^e (1 - \ln x)^3 dx$ 2

- (iii) Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$ where $n = 1, 2, 3, \dots$ 2

- (iv) Show that $0 \leq I_n \leq e - 1$ 1

- (v) Deduce that $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = e$ 1

End of paper.

Student Number



2016

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

Multiple-Choice Answer Sheet

Select the alternative A, B, C, or D that best answers the question by placing a **X** in the box.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Mathematics

Factorisation

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Angle sum of a polygon

$$S = (n - 2) \times 180^\circ$$

Equation of a circle

$$(x - h)^2 + (y - k)^2 = r^2$$

Trigonometric ratios and identities

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

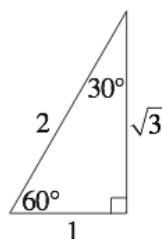
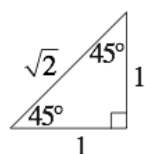
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

Exact ratios



Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C$$

Distance between two points

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Perpendicular distance of a point from a line

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

Slope (gradient) of a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form of the equation of a line

$$y - y_1 = m(x - x_1)$$

n th term of an arithmetic series

$$T_n = a + (n - 1)d$$

Sum to n terms of an arithmetic series

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \text{or} \quad S_n = \frac{n}{2} (a + l)$$

n th term of a geometric series

$$T_n = ar^{n-1}$$

Sum to n terms of a geometric series

$$S_n = \frac{a(r^n - 1)}{r - 1} \quad \text{or} \quad S_n = \frac{a(1 - r^n)}{1 - r}$$

Limiting sum of a geometric series

$$S = \frac{a}{1 - r}$$

Compound interest

$$A_n = P \left(1 + \frac{r}{100} \right)^n$$

Mathematics (continued)

Differentiation from first principles

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Derivatives

If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$

If $y = uv$, then $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

If $y = \frac{u}{v}$, then $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

If $y = F(u)$, then $\frac{dy}{dx} = F'(u) \frac{du}{dx}$

If $y = e^{f(x)}$, then $\frac{dy}{dx} = f'(x)e^{f(x)}$

If $y = \log_e f(x) = \ln f(x)$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$

If $y = \sin f(x)$, then $\frac{dy}{dx} = f'(x) \cos f(x)$

If $y = \cos f(x)$, then $\frac{dy}{dx} = -f'(x) \sin f(x)$

If $y = \tan f(x)$, then $\frac{dy}{dx} = f'(x) \sec^2 f(x)$

Solution of a quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum and product of roots of a quadratic equation

$$\alpha + \beta = -\frac{b}{a} \qquad \alpha\beta = \frac{c}{a}$$

Equation of a parabola

$$(x-h)^2 = \pm 4a(y-k)$$

Integrals

$$\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int e^{ax+b} dx = \frac{1}{a} e^{ax+b} + C$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

$$\int \sin(ax+b) dx = -\frac{1}{a} \cos(ax+b) + C$$

$$\int \cos(ax+b) dx = \frac{1}{a} \sin(ax+b) + C$$

$$\int \sec^2(ax+b) dx = \frac{1}{a} \tan(ax+b) + C$$

Trapezoidal rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]$$

Simpson's rule (one application)

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

Logarithms – change of base

$$\log_a x = \frac{\log_b x}{\log_b a}$$

Angle measure

$$180^\circ = \pi \text{ radians}$$

Length of an arc

$$l = r\theta$$

Area of a sector

$$\text{Area} = \frac{1}{2} r^2 \theta$$

Mathematics Extension 1

Angle sum identities

$$\sin(\theta + \phi) = \sin\theta \cos\phi + \cos\theta \sin\phi$$

$$\cos(\theta + \phi) = \cos\theta \cos\phi - \sin\theta \sin\phi$$

$$\tan(\theta + \phi) = \frac{\tan\theta + \tan\phi}{1 - \tan\theta \tan\phi}$$

t formulae

If $t = \tan \frac{\theta}{2}$, then

$$\sin\theta = \frac{2t}{1+t^2}$$

$$\cos\theta = \frac{1-t^2}{1+t^2}$$

$$\tan\theta = \frac{2t}{1-t^2}$$

General solution of trigonometric equations

$$\sin\theta = a, \quad \theta = n\pi + (-1)^n \sin^{-1}a$$

$$\cos\theta = a, \quad \theta = 2n\pi \pm \cos^{-1}a$$

$$\tan\theta = a, \quad \theta = n\pi + \tan^{-1}a$$

Division of an interval in a given ratio

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

Parametric representation of a parabola

For $x^2 = 4ay$,

$$x = 2at, \quad y = at^2$$

At $(2at, at^2)$,

tangent: $y = tx - at^2$

normal: $x + ty = at^3 + 2at$

At (x_1, y_1) ,

tangent: $xx_1 = 2a(y + y_1)$

normal: $y - y_1 = -\frac{2a}{x_1}(x - x_1)$

Chord of contact from (x_0, y_0) : $xx_0 = 2a(y + y_0)$

Acceleration

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

Simple harmonic motion

$$x = b + a \cos(nt + \alpha)$$

$$\ddot{x} = -n^2(x - b)$$

Further integrals

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Sum and product of roots of a cubic equation

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

Estimation of roots of a polynomial equation

Newton's method

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

Solutions

$$Q1 \quad v = r\omega \quad 40\text{revs} = \frac{40 \times 2\pi}{60} = \frac{4\pi}{3}$$

$$r \approx 30 \div \frac{4\pi}{3} \approx 7 \quad \therefore C$$

$$Q2 \quad 4x^2 - 25y^2 = 9 \Rightarrow a^2 = \frac{9}{4}, b^2 = \frac{9}{25}$$

$$b^2 = a^2(e^2 - 1) \Rightarrow \frac{9}{25} = \frac{9}{4}(e^2 - 1)$$

$$4 = 25e^2 - 25 \Rightarrow e^2 = \frac{29}{25} \Rightarrow e = \frac{\sqrt{29}}{5} \therefore B$$

$$Q3 \quad z = 3 - i \Rightarrow iz = 1 + 3i \Rightarrow \bar{iz} = 1 - 3i \therefore C$$

$$Q4 \quad y = \pm \sqrt{f(x)} \quad \text{only one graph has symmetry about x-axis}$$

$$\therefore C$$

$$Q5 \quad r = 2 - x \Rightarrow V = \pi \int_{-1}^1 [(2-x)^2 - 1^2] dy = \pi \int_{-1}^1 (x^2 - 4x + 3) dy$$

$$= \pi \int_{-1}^1 (3 + 4y^2 + y^4) dy \therefore B$$

$$Q6 \quad \ddot{x} = -g + kv^2 \Rightarrow v \frac{dv}{dx} = -g + kv^2 \Rightarrow \frac{dx}{dv} = \frac{v}{kv^2 - g}$$

$$\therefore x = \frac{1}{2k} \ln |kv^2 - g| + c$$

$$v = 0, x = 0 \therefore c = -\frac{1}{2k} \ln g$$

$$-2kx = \ln \left| \frac{g - kv^2}{g} \right| \Rightarrow g \times e^{-2kx} = g - kv^2$$

$$kv^2 = g - g \times e^{-2kx} \Rightarrow v^2 = \frac{g}{k} (1 - e^{-2kx}) \therefore A$$

$$Q7 \quad S \text{ on } y \text{ axis } \therefore \text{not } C$$

$$\frac{x^2}{1} + \frac{y^2}{8} = 1 \Rightarrow e^2 = 1 - \frac{a^2}{b^2} = \frac{7}{8} \Rightarrow be \neq 3$$

$$\frac{x^2}{4} + \frac{y^2}{5} = 1 \Rightarrow e^2 = 1 - \frac{a^2}{b^2} = \frac{1}{5} \Rightarrow be \neq 3$$

$$\frac{x^2}{16} + \frac{y^2}{25} = 1 \Rightarrow e^2 = 1 - \frac{a^2}{b^2} = \frac{9}{25} \Rightarrow be = 5 \times \frac{3}{5} = 3 \therefore D$$

	A	B	C	D
1			X	
2		X		
3			X	
4			X	
5		X		
6	X			
7				X
8		X		
9			X	
10		X		

$$\begin{aligned} \text{Q8} \quad \sum \alpha^3 - 2 \sum \alpha^2 + 9 &= 0 \\ \sum \alpha^3 &= 2 \sum \alpha^2 - 9 = 2 \left[\left(\sum \alpha \right)^2 - 2 \sum \alpha \beta \right] - 9 \\ &= 2[4 - 0] - 9 = -1 \quad \therefore B \end{aligned}$$

$$\begin{aligned} \text{Q9} \quad y^2 - 4y + x &= 0 \Rightarrow y = \frac{4 \pm \sqrt{16 - 4x}}{2} = 2 \pm \sqrt{4 - x} \\ \therefore y_1 - y_2 &= 2\sqrt{4 - x} \Rightarrow A = \sqrt{4 - x} \times 2\sqrt{4 - x} \\ \Rightarrow \delta V &= 2(4 - x) \delta x \Rightarrow V = \int_0^4 2(4 - x) dx \quad \therefore C \end{aligned}$$

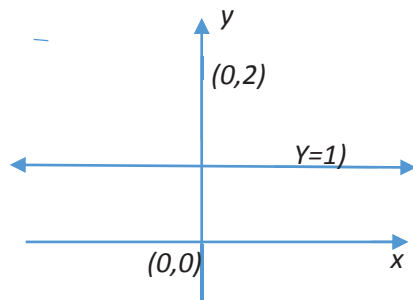
$$\begin{aligned} \text{Q10} \quad y &= \frac{c^2}{x} \Rightarrow y' = -\frac{c^2}{x^2} @ x = cp \Rightarrow y' = m_{\text{tang}} = \frac{-1}{p^2} \Rightarrow m_{\text{norm}} = p^2 \\ \text{Normal is } y - \frac{c}{p} &= p^2(x - cp) \Rightarrow p^2 x - y - cp^3 + \frac{c}{p} \\ \Rightarrow p^3 x - py + c - cp^4 &= 0 \quad \therefore B \end{aligned}$$

$$11a)i) \quad z = (1 - i)^{-1} = \frac{1}{1 - i} \times \frac{1 + i}{1 + i} = \frac{1}{2} + \frac{1}{2}i$$

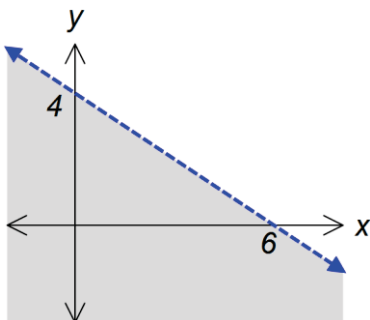
$$\therefore \bar{z} = \frac{1}{2} - \frac{1}{2}i = \frac{1}{\sqrt{2}} \text{cis} \left(-\frac{\pi}{4} \right)$$

$$ii) \quad (\bar{z})^{13} = \frac{1}{64\sqrt{2}} \text{cis} \left(-\frac{13\pi}{4} \right) = \frac{1}{64\sqrt{2}} \text{cis} \left(\frac{3\pi}{4} \right) = \frac{1}{64\sqrt{2}} \left(-\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right) = \frac{-1}{128} + i \frac{1}{128}$$

$$\begin{aligned} b) \quad |z - 2i| &= |z| \Rightarrow \text{locus is all points equidistant from } (0, 0) \text{ and } (0, 2) \\ \text{ie } y &= 1 \end{aligned}$$



$$\begin{aligned} c) \quad \Re[(2 - 3i)z] &< 12 \\ (2 - 3i)(x + iy) &= 2x + 3y + i(2y - 3x) \\ \therefore 2x + 3y &< 12 \end{aligned}$$



d) $x^3 - 3x^2 - x + 2 = 0 \Rightarrow \sum \alpha = 3$
 $\therefore 2\alpha + \beta + \gamma = \alpha + \sum \alpha = \alpha + 3$ and similarly for the other roots
 $\therefore$ replace x with $y-3 \Rightarrow (y-3)^3 - 3(y-3)^2 - (y-3) + 2 = 0$
 $\Rightarrow y^3 - 9y^2 + 27y - 27 - 3y^2 + 18y - 27 - y + 3 + 2 = 0$
 $\Rightarrow y^3 - 12y^2 + 44y - 49 = 0$

e) $\frac{x^2 + x + 2}{(x^2 + 1)(x + 1)} = \frac{Ax + B}{x^2 + 1} + \frac{C}{x + 1} \Rightarrow x^2 + x + 2 = (Ax + B)(x + 1) + C(x^2 + 1)$

let $x = -1 \Rightarrow 2 = 2C \therefore C = 1$

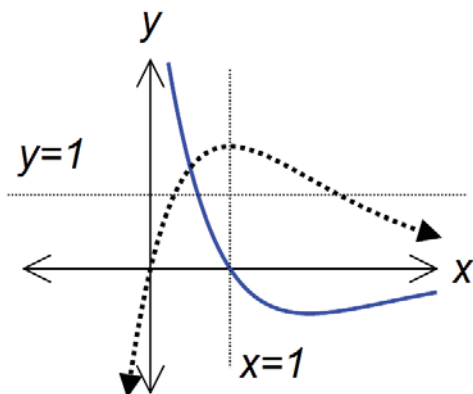
let $x = 0 \Rightarrow 2 = B + C \Rightarrow B = 1$

coefficients of $x^2 \Rightarrow 1 = A + C \therefore A = 0$

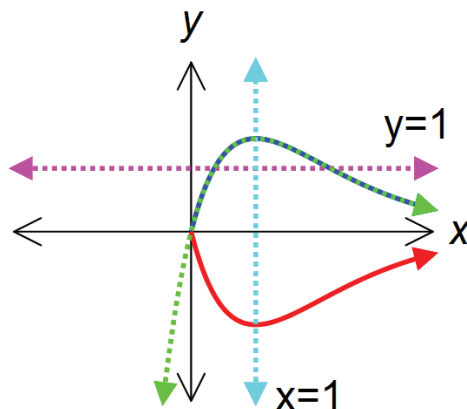
$$\frac{x^2 + x + 2}{(x^2 + 1)(x + 1)} = \left(\frac{1}{x^2 + 1} + \frac{1}{x + 1} \right)$$

ii) $\int \frac{x^2 + x + 2}{(x^2 + 1)(x + 1)} dx = \int \left(\frac{1}{x^2 + 1} + \frac{1}{x + 1} \right) dx = \tan^{-1} x + \ln|x + 1| + d$

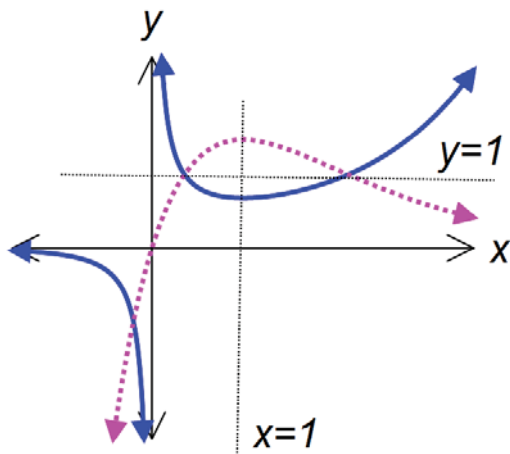
12a)i) $y = f'(x)$



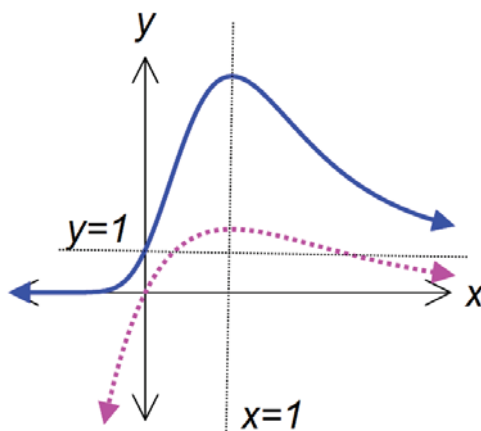
ii) $|y| = f(x)$



iii) $y = \frac{1}{f(x)}$



iv) $y = 3^{f(x)}$



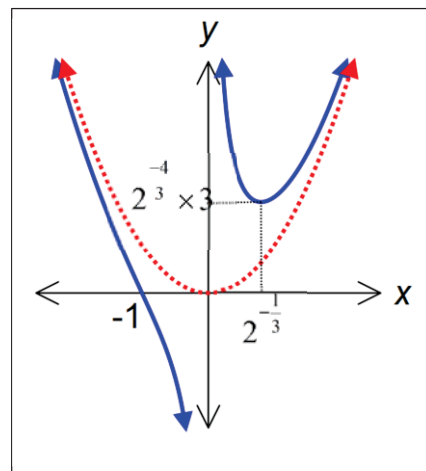
$$b) \int_4^7 \frac{dx}{x^2 - 8x + 19} = \int_4^7 \frac{dx}{(x-4)^2 + 3} = \frac{1}{\sqrt{3}} \left[\tan^{-1} \left(\frac{x-4}{\sqrt{3}} \right) \right]_4^7 = \frac{1}{\sqrt{3}} \left[\tan^{-1} \sqrt{3} - \tan^{-1} 0 \right] = \frac{\pi}{3\sqrt{3}}$$

$$c)i) f(x) = \frac{x^3 + 1}{x} = x^2 + \frac{1}{x} \Rightarrow f(x) - x^2 = \frac{1}{x} \therefore \lim_{x \rightarrow \infty} (f(x) - x^2) = \lim_{x \rightarrow \infty} \left(\frac{1}{x} \right) = 0$$

$$ii) f'(x) = 2x - \frac{1}{x^2} = 0 \text{ when } 2x^3 = 1 \Rightarrow x = 2^{-\frac{1}{3}}$$

$$\text{also } x \rightarrow 0^+ f(x) \rightarrow \infty, \quad x \rightarrow 0^- f(x) \rightarrow -\infty$$

$$\text{if } f(x) = 0 \Rightarrow x^3 = -1 \Rightarrow x = -1$$



$$13a) z^5 - 1 = 0 \Rightarrow (z-1)(z^4 + z^3 + z^2 + z + 1)$$

$$\text{if } \omega \text{ is complex } z-1 \neq 0 \therefore z^4 + z^3 + z^2 + z + 1 = 0$$

$$\text{let } \alpha = \omega + \omega^4, \quad \beta = \omega^2 + \omega^3 \therefore \alpha + \beta = -1$$

$$\alpha\beta = (\omega + \omega^4)(\omega^2 + \omega^3) = \omega^3 + \omega^4 + \omega + \omega^2 = -1$$

$$\therefore \text{quadratic is } z^2 - (\alpha + \beta)z + \alpha\beta = z^2 + z - 1 = 0$$

$$b)i) P(x) = x^3 + x^2 + mx + n \quad \text{since coefficients real if } z = 1 - 2i \text{ is a root so too is } z = 1 + 2i$$

$$\therefore \sum \alpha = 1 - 2i + 1 + 2i + \alpha = -1 \Rightarrow \alpha = -3$$

$$\therefore P(x) = (x+3)(x-1-2i)(x-1+2i)$$

$$ii) P(-3) = -27 + 9 - 3m + n = 0 \Rightarrow 3m - n = -18 \quad (1)$$

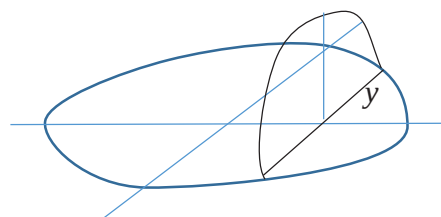
$$\prod \alpha = -3(1-2i)(1+2i) = -15 = -n \Rightarrow n = 15$$

$$\text{sub in (1)} \Rightarrow 3m = -18 + 15 \Rightarrow m = -1$$

$$c)i) \text{ since } a = 6, b = 4 \Rightarrow \frac{x^2}{36} + \frac{y^2}{16} = 1$$

$$ii) A = \frac{\pi}{2} y^2 \Rightarrow \delta V = \frac{\pi}{2} y^2 \delta x$$

$$\begin{aligned} \therefore V &= \lim_{\delta x \rightarrow 0} \sum_{x=-6}^6 \left(\frac{\pi}{2} y^2 \delta x \right) = 2 \times \frac{\pi}{2} \int_0^6 16 \left(1 - \frac{x^2}{36} \right) dx \\ &= 16\pi \left[x - \frac{x^3}{108} \right]_0^6 = 16\pi \left(6 - \frac{216}{108} \right) = 64\pi \text{ U}^3 \end{aligned}$$

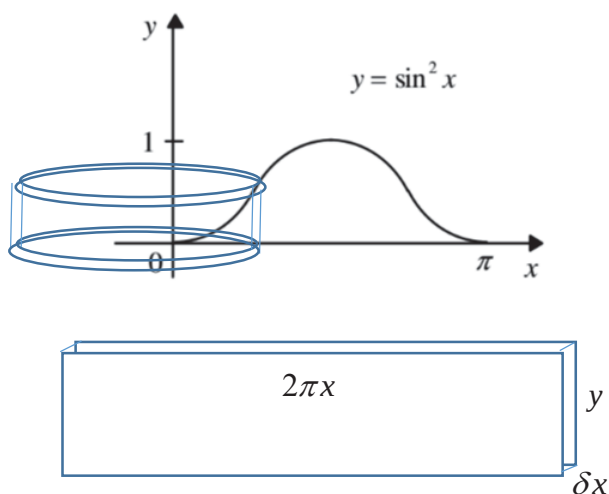


d)i) let $P(x) = (x - \alpha)^3 (x - \beta)$
 $\therefore P'(x) = (x - \alpha)^3 \times 1 + 3(x - \alpha)^2 (x - \beta)$
 $= (x - \alpha)^2 [x - \alpha + 3x - 3\beta]$
 $\therefore P'(x)$ has a zero of multiplicity 2

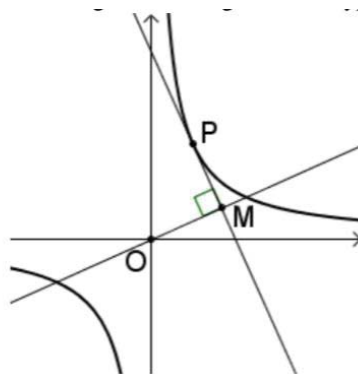
ii) $P(x) = 8x^4 - 25x^3 + 27x^2 - 11x + 1$
 $P'(x) = 32x^3 - 75x^2 + 54x - 11$
 $P''(x) = 96x^2 - 150x + 54$
 $P''(1) = 96 - 150 + 54 = 0$ and $P'(1) = 32 - 75 + 54 - 11 = 0$
 $\therefore x = 1$ is a triple root $\Rightarrow P(x) = (x - 1)^3 (8x - 1)$
 $\therefore$ zeros are $1, 1, 1, \frac{1}{8}$

14a)i) $\int_0^a f(a - x) dx = \int_0^a f(x) dx \therefore \int_0^\pi x \cos 2x dx = \int_0^\pi (\pi - x) \cos(2\pi - 2x) dx = \int_0^\pi (\pi - x) \cos(2x) dx$
 $\int_0^\pi x \cos 2x dx = \int_0^\pi \pi \cos 2x dx - \int_0^\pi x \cos 2x dx \Rightarrow 2I = \pi \left[\frac{1}{2} \sin 2x \right]_0^\pi = 0 \therefore I = 0$

a)ii) $\delta V = \pi [(x + \delta x)^2 - x^2] y$
 $\approx 2\pi xy \delta x$
 $V \approx \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi} \delta V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi} 2\pi xy \delta x$
 $= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\pi} 2\pi x \sin^2 x \delta x$
 $\therefore V = \pi \int_0^\pi x (1 - \cos 2x) dx$
 $= \pi \int_0^\pi x dx - \pi \int_0^\pi x \cos 2x dx$
 $= \pi \left[\frac{x^2}{2} \right]_0^\pi - 0 \text{ from ii)}$
 $= \frac{\pi^3}{2}$



b)i) $xy = 1 \Rightarrow y = \frac{1}{x} \Rightarrow y' = \frac{-1}{x^2} = \frac{-1}{t^2}$ at $x = t$
 $\therefore$ tangent is $y - \frac{1}{t} = \frac{-1}{t^2} (x - t)$
 $\Rightarrow x + t^2 y = 2t \quad (1)$
ii) equation of OM $y - 0 = t^2 (x - 0) \Rightarrow y = t^2 x$



iii) coordinates of M solve (1) with (2)

$$\text{sub } y = t^2 x \text{ in } x + t^2 y = 2t$$

$$x + t^4 x = 2t \Rightarrow x = \frac{2t}{1+t^4}, y = \frac{2t^3}{1+t^4}$$

$$\text{from (2) } \frac{y}{x} = t^2 \Rightarrow t = \pm \sqrt{\frac{y}{x}} \quad \therefore x = \frac{2\left(\pm \sqrt{\frac{y}{x}}\right)}{1 + \frac{y^2}{x^2}} = \frac{2x^2\left(\pm \sqrt{\frac{y}{x}}\right)}{x^2 + y^2}$$

$$x(x^2 + y^2) = 2x^2\left(\pm \sqrt{\frac{y}{x}}\right) \Rightarrow x^2(x^2 + y^2)^2 = 4x^4 \frac{y}{x}$$

$$x^2(x^4 + 2x^2y^2 + y^4 - 4xy) = 0 \text{ reject } x^2 = 0$$

$$\therefore x^4 + 2x^2y^2 + y^4 - 4xy = 0$$

Restrictions: $\frac{y}{x} > 0$ for t to exist and $xy < 1$ for M to lie outside H

c) $\ddot{x} = -g - kv \Rightarrow v \frac{dv}{dx} = -g - kv$

$$\frac{v}{g + kv} dv = -dx \Rightarrow \frac{\frac{1}{k}(kv + g) - \frac{g}{k}}{g + kv} = -dx$$

$$\int_v^0 \frac{1}{k} dv - \frac{g}{k} \int_v^0 \frac{1}{g + kv} dv = \int_0^H dx$$

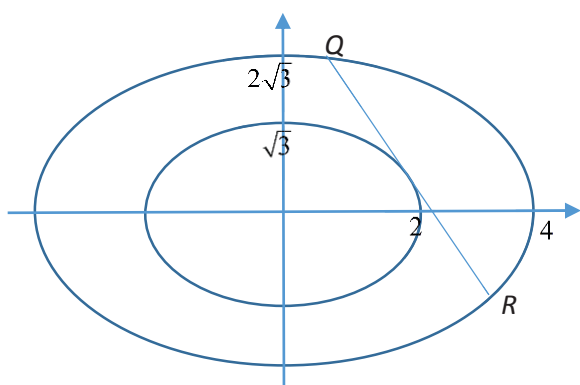
$$\left[\frac{v}{k} - \frac{g}{k^2} \ln(g + kv) \right]_v^0 = -H$$

$$-\frac{g}{k^2} \ln g - \frac{V}{k} + \frac{g}{k^2} \ln(g + kV) = -H$$

$$H = \frac{V}{k} - \frac{g}{k^2} \ln\left(\frac{g + kV}{g}\right) = \frac{V}{k} - \frac{g}{k^2} \ln\left(1 + \frac{kV}{g}\right)$$

15a) $\int_1^e x^7 \ln x \, dx = \int_1^e \ln x \frac{d}{dx} \left(\frac{x^8}{8} \right) dx = \left[\frac{x^8}{8} \ln x \right]_1^e - \int_1^e \frac{x^8}{8} \times \frac{1}{x} dx$
 $= \frac{e^8}{8} - \frac{1}{8} \int_1^e x^7 \, dx = \frac{e^8}{8} - \frac{e^8}{64} + \frac{1}{64} = \frac{7e^8 + 1}{64}$

b)



i) $b^2 = a^2(1 - e^2) \Rightarrow \frac{3}{4} - 1 = -e^2$

$$\Rightarrow e^2 = \frac{1}{4} \Rightarrow e = \frac{1}{2}$$

$$\therefore \text{Foci are } (\pm ae, 0), S, S' = (\pm 1, 0)$$

$$\text{directrices are } x = \pm \frac{a}{e} = \pm \frac{2}{\frac{1}{2}} = \pm 4$$

$$ii) \quad \frac{x^2}{4} + \frac{y^2}{3} = 1 \Rightarrow \frac{x}{2} + \frac{2y}{3} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-3x}{4y} = \frac{-6 \cos p}{4\sqrt{3} \sin p} = -\frac{\sqrt{3} \cos p}{2 \sin p} @ P$$

$$\therefore \text{Tangent at } P \text{ is } y - \sqrt{3} \sin p = -\frac{\sqrt{3} \cos p}{2 \sin p} (x - 2 \cos p)$$

$$2y \sin p - 2\sqrt{3} \sin^2 p = -\sqrt{3} x \cos p + 2\sqrt{3} \cos^2 p$$

$$\sqrt{3} x \cos p + 2y \sin p = 2\sqrt{3} \Rightarrow \frac{x \cos p}{2} + \frac{y \sin p}{\sqrt{3}} = 1$$

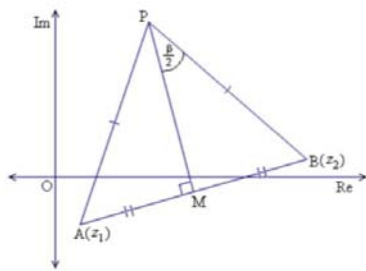
$$iii) \quad \text{Let this tangent meet } E_2 \text{ at } T (4 \cos t, 2\sqrt{3} \sin t)$$

$$\therefore \frac{4 \cos t \cos p}{2} + \frac{2\sqrt{3} \sin t \sin p}{\sqrt{3}} = 1 \Rightarrow \cos t \cos p + \sin t \sin p = \frac{1}{2}$$

$$\therefore \cos(t - p) = \frac{1}{2} \Rightarrow t - p = \pm \frac{\pi}{3} \text{ since } 0 < p < \frac{\pi}{2} \text{ and } -\pi < t < \pi$$

$$\therefore \text{if } q = p + \frac{\pi}{3} \text{ then } r = p - \frac{\pi}{3} \Rightarrow |q - r| = \frac{2\pi}{3}$$

c)



$$c)i)\alpha) \quad \overrightarrow{OA} + \overrightarrow{AM} = \overrightarrow{OM}$$

$$\text{or } \overrightarrow{AM} = \frac{1}{2} \overrightarrow{AB} = \frac{1}{2} (z_2 - z_1)$$

$$\overrightarrow{AM} = \overrightarrow{OM} - \overrightarrow{OA}$$

$$= \frac{z_1 + z_2}{2} - z_1 = \frac{z_2 - z_1}{2}$$

$$\beta) \quad \overrightarrow{MP} = \overrightarrow{AM} \text{ rotated through } 90^\circ \text{ and increased in length by a factor } \cot \frac{\beta}{2}$$

$$\left(\text{since } \tan \frac{\beta}{2} = \frac{|AM|}{|PM|} \Rightarrow |PM| = \cot \frac{\beta}{2} |AM| \right)$$

$$\therefore \overrightarrow{MP} = i \left(\frac{z_2 - z_1}{2} \right) \cot \frac{\beta}{2}$$

$$ii) \quad \overrightarrow{OP} = \overrightarrow{OM} + \overrightarrow{MP} = \frac{z_1 + z_2}{2} + i \left(\frac{z_2 - z_1}{2} \right) \cot \frac{\beta}{2}$$

$$= \frac{1}{2} z_1 \left[1 - i \cot \frac{\beta}{2} \right] + \frac{1}{2} z_2 \left[1 + i \cot \frac{\beta}{2} \right]$$

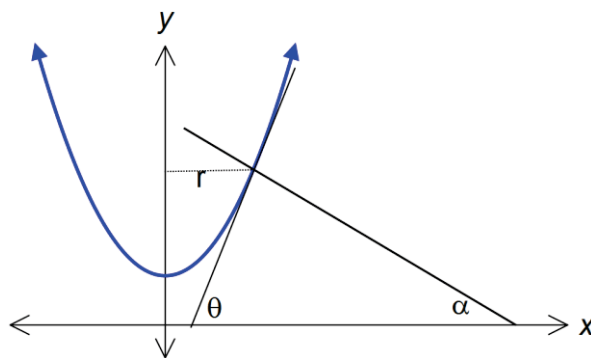
$$16a)i) \quad y^2 - x^2 = 1 \Rightarrow y = \sqrt{1+x^2}$$

$$\Rightarrow y' = \frac{1}{2}(1+x^2)^{-\frac{1}{2}} \times 2x = \frac{x}{\sqrt{1+x^2}}$$

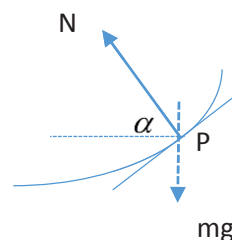
$$\text{when } x=r \quad y' = \tan \theta = \frac{r}{\sqrt{1+r^2}}$$

$$\text{but from diagram } \alpha = \frac{\pi}{2} - \theta$$

$$\therefore \tan \alpha = \cot \theta = \frac{\sqrt{1+r^2}}{r}$$



ii)



iii) let the normal reaction be N
resolve forces vertically and horizontally

$$V: N \sin \alpha = mg \quad (1)$$

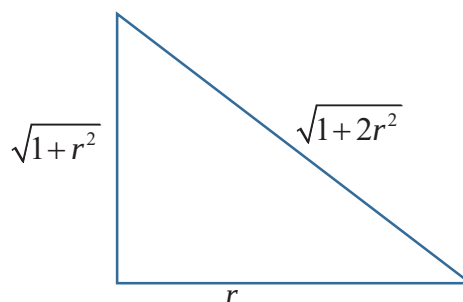
$$H: N \cos \alpha = m r \omega^2 \quad (2)$$

$$(1) \div (2) \Rightarrow \tan \alpha = \frac{g}{r \omega^2} = \frac{\sqrt{1+r^2}}{r}$$

$$\therefore \frac{g}{\omega^2} = \sqrt{1+r^2} \Rightarrow r^2 = \frac{g^2}{\omega^4} - 1$$

$$\text{sub in (1)} \Rightarrow N = \frac{mg}{\sin \alpha} = mg \div \frac{\sqrt{1+r^2}}{\sqrt{1+2r^2}} = mg \div \frac{\frac{g}{\omega^2}}{\sqrt{1+2\left(\frac{g^2}{\omega^4}-1\right)}}$$

$$= mg \times \frac{\sqrt{\frac{2g^2}{\omega^4}-1}}{\frac{g}{\omega^2}} = m \sqrt{2g^2 - \omega^4}$$



$$iv) \quad \text{from } H \quad y = \sqrt{1+x^2} \quad \text{when } x=r \Rightarrow y = \sqrt{1+r^2}$$

$$\text{from iii) } r^2 = \frac{g^2}{\omega^4} - 1 \Rightarrow y = \sqrt{\frac{g^2}{\omega^4}} = \frac{g}{\omega^2}$$

$$\text{but } 1 \leq y \leq 5 \quad \therefore \quad 1 \leq \frac{g}{\omega^2} \leq 5 \Rightarrow \frac{1}{5} \leq \frac{\omega^2}{g} \leq 1$$

$$\Rightarrow \sqrt{\frac{g}{5}} \leq \omega \leq \sqrt{g}$$

$$16(b) \quad \text{Let } I_n = \int_1^e (1 - \ln x)^n dx \quad \text{where } n = 0, 1, 2, \dots$$

$$(i) \quad \text{Show that } I_n = -1 + n I_{n-1} \quad \text{where } n = 0, 1, 2, \dots$$

$$\begin{aligned}
I_n &= \int_1^e (1 - \ln x)^n dx = \int_1^e (1 - \ln x)^n \frac{d}{dx}(x) dx \\
&= \left[x(1 - \ln x)^n \right]_1^e - \int_1^e nx(1 - \ln x)^{n-1} \times \frac{-1}{x} dx \\
&= -1 + nI_{n-1}
\end{aligned}$$

(ii) Hence evaluate $\int_1^e (1 - \ln x)^3 dx$

2

$$\begin{aligned}
I_3 &= -1 + 3I_2 = -1 + 3(-1 + 2I_1) = -1 - 3 + 6(-1 + I_0) \\
&= -10 + 6 \int_1^e dx = -10 + 6e - 6 = 6e - 16
\end{aligned}$$

(iii) Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$ where $n = 1, 2, 3, \dots$

2

$$\begin{aligned}
\frac{I_n}{n!} &= \frac{-1 + nI_{n-1}}{n!} = \frac{-1}{n!} + \frac{I_{n-1}}{(n-1)!} = \frac{-1}{n!} + \frac{-1 + (n-1)I_{n-2}}{(n-1)!} = \frac{-1}{n!} + \frac{-1}{(n-1)!} + \frac{I_{n-2}}{(n-2)!} \\
&= \frac{-1}{n!} + \frac{-1}{(n-1)!} + \frac{-1 + (n-2)I_{n-3}}{(n-2)!} = \frac{-1}{n!} + \frac{-1}{(n-1)!} + \frac{-1}{(n-2)!} + \frac{I_{n-3}}{(n-3)!} \\
&= \frac{-1}{n!} + \frac{-1}{(n-1)!} + \frac{-1}{(n-2)!} + \dots + \frac{-1}{1!} + \frac{I_0}{0!} \\
&= -\sum_{r=1}^n \frac{1}{r!} + \int_1^e dx = -\sum_{r=1}^n \frac{1}{r!} + [x]_1^e = -\sum_{r=1}^n \frac{1}{r!} + e - 1 = -\sum_{r=1}^n \frac{1}{r!} + e - \frac{1}{0!} = -\sum_{r=0}^n \frac{1}{r!} + e
\end{aligned}$$

(iv) Show that $0 \leq I_n \leq e - 1$

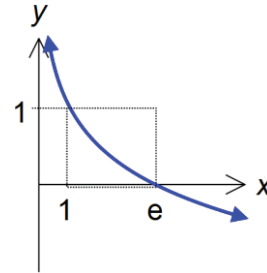
1

Consider $y = 1 - \ln x$ for $1 \leq x \leq e \Rightarrow 0 \leq y \leq 1$

$\therefore$ for $y = (1 - \ln x)^n$ $0 \leq y \leq 1$ also

clearly $\int_1^e (1 - \ln x)^n dx \geq 0$ but $\leq$ rectangle $= 1 \times (e - 1)$

$$\therefore 0 \leq I_n \leq (e - 1)$$



(v) Deduce that $\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{1}{r!} = e$

1

from (iv) $0 \leq \frac{I_n}{n!} \leq \frac{e-1}{n!}$

$$\therefore n \rightarrow \infty \quad \frac{e-1}{n!} \rightarrow 0 \quad \therefore 0 \leq \lim_{n \rightarrow \infty} \frac{I_n}{n!} \leq 0$$

$$\text{using (iii)} \quad \lim_{n \rightarrow \infty} \left(e - \sum_{r=0}^n \frac{1}{r!} \right) = 0 \Rightarrow e - \lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \frac{1}{r!} \right) = 0 \Rightarrow e = \lim_{n \rightarrow \infty} \left(\sum_{r=0}^n \frac{1}{r!} \right)$$